

## UNIT 4

**Algebraic Structures:** Definition, Groups, Subgroups and order, Cyclic Groups, Cosets, Lagrange's theorem, Normal Subgroups, Permutation and Symmetric groups, Group Homomorphisms, Definition and elementary properties of Rings and Fields.

# Algebraic Structures

## 1. Algebraic Structure

### Definition

An **algebraic structure** consists of a non-empty set along with one or more binary operations defined on it that satisfy certain axioms.

### Example

- $(\mathbb{Z}, +)$  where  $\mathbb{Z}$  is the set of integers and  $+$  is addition
- $(\mathbb{N}, \times)$  where  $\mathbb{N}$  is the set of natural numbers and  $\times$  is multiplication

## 2. Group

### Definition

A **group** is a set  $G$  together with a binary operation  $*$  satisfying the following axioms:

| Property      | Description                                                                 |
|---------------|-----------------------------------------------------------------------------|
| Closure       | For all $a, b \in G$ , $a * b \in G$                                        |
| Associativity | $(a * b) * c = a * (b * c)$                                                 |
| Identity      | There exists $e \in G$ such that $a * e = e * a = a$                        |
| Inverse       | For each $a \in G$ , there exists $a^{-1} \in G$ such that $a * a^{-1} = e$ |

### Example

$(\mathbb{Z}, +)$

- Identity element: 0
- Inverse of  $a$ :  $-a$

Hence,  $(\mathbb{Z}, +)$  is a group.

## 3. Subgroup

### Definition

A **subgroup**  $H$  of a group  $G$  is a non-empty subset of  $G$  that itself forms a group under the same operation.

## Subgroup Test

A non-empty subset  $H \subseteq G$  is a subgroup if:

For all  $a, b \in H$ ,  $a * b^{-1} \in H$

## Example

Let  $G = (\mathbb{Z}, +)$

$H = \{\dots, -4, -2, 0, 2, 4, \dots\}$

$H$  is a subgroup of  $\mathbb{Z}$  under addition.

## 4. Order

### Order of a Group

The **order of a group** is the total number of elements in the group.

It is denoted by  $|G|$ .

### Order of an Element

The **order of an element  $a$**  in group  $G$  is the smallest positive integer  $n$  such that:

$$a^n = e$$

## Example

In  $\mathbb{Z}_6$  under addition, the order of element 2 is 3.

## 5. Cyclic Group

### Definition

A group  $G$  is called **cyclic** if there exists an element  $a \in G$  such that every element of  $G$  can be written as a power of  $a$ .

$$G = \langle a \rangle$$

## Example

$(\mathbb{Z}, +)$  is a cyclic group generated by 1.

### Properties of Cyclic Groups

#### Property

Every cyclic group is abelian

Every subgroup of a cyclic group is cyclic

Finite cyclic groups are isomorphic to  $\mathbb{Z}_n$

## 6. Cosets

### Definition

Let  $H$  be a subgroup of  $G$  and  $a \in G$ .

| Type        | Definition       |
|-------------|------------------|
| Left Coset  | $aH = \{a * h\}$ |
| Right Coset | $Ha = \{h * a\}$ |

### Example

Let  $G = \mathbb{Z}$  and  $H = 2\mathbb{Z}$

| Coset   | Elements      |
|---------|---------------|
| $0 + H$ | Even integers |
| $1 + H$ | Odd integers  |

## 7. Lagrange's Theorem

### Statement

If  $H$  is a subgroup of a finite group  $G$ , then:

**$|H|$  divides  $|G|$**

### Consequences

| Result                                      |
|---------------------------------------------|
| Order of any element divides order of group |
| Order of subgroup divides order of group    |

### Example

If  $|G| = 12$ , then possible subgroup orders are:

1, 2, 3, 4, 6, 12

## 8. Normal Subgroup

### Definition

A subgroup  $N$  of  $G$  is called a **normal subgroup** if:

$aN = Na$  for all  $a \in G$

It is denoted as  **$N \triangleleft G$**

### Example

In an abelian group, every subgroup is normal.

## 9. Permutation Group

### Definition

A **permutation** is a one-to-one and onto mapping of a set onto itself.

### Example

For the set  $\{1, 2, 3\}$ , one permutation is:

$(1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 1)$

### Permutation Group

The set of all permutations of a set forms a group under composition of functions.

## 10. Symmetric Group

### Definition

The **symmetric group  $S_n$**  is the group of all permutations of  $n$  elements.

### Properties

| Property                            |
|-------------------------------------|
| Order of $S_n$ is $n!$              |
| $S_n$ is non-abelian for $n \geq 3$ |

### Example

$S_3$  has 6 elements.

## 11. Group Homomorphism

### Definition

A function  $f : G \rightarrow G'$  is a **group homomorphism** if:

$$f(a * b) = f(a) * f(b)$$

### Example

$$f : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$f(x) = x \bmod 6$$

### Properties

| Property                  |
|---------------------------|
| $f(e) = e'$               |
| $f(a^{-1}) = (f(a))^{-1}$ |

## 12. Rings

### Definition

A **ring** is a non-empty set  $R$  with two binary operations  $+$  and  $\times$  such that:

| Condition                     |
|-------------------------------|
| $(R, +)$ is an abelian group  |
| Multiplication is associative |
| Distributive laws hold        |

### Example

$(\mathbb{Z}, +, \times)$  is a ring.

## 13. Fields

### Definition

A **field** is a ring in which every non-zero element has a multiplicative inverse and multiplication is commutative.

### Examples

- Rational numbers ( $\mathbb{Q}$ )
- Real numbers ( $\mathbb{R}$ )
- $\mathbb{Z}_p$  where  $p$  is a prime number

## Difference Between Ring and Field

| Feature                    | Ring        | Field            |
|----------------------------|-------------|------------------|
| Additive identity          | Yes         | Yes              |
| Additive inverse           | Yes         | Yes              |
| Multiplicative identity    | Optional    | Mandatory        |
| Multiplicative inverse     | Not for all | For all non-zero |
| Commutative multiplication | Optional    | Yes              |